

Emotional Models: Supervised vs. Zero-Shot Classification

Miguel Castela¹ and Miguel Martins²

Univsersity of Coimbra, Portugal
uc2022212972@student.uc.pt
uc2022213951@student.uc.pt

1 Introduction

The core task addressed in this work is multi-class emotion classification. Formally, given an input text sequence x (e.g., a tweet or a short message), the goal is to map x to a single label y from a predefined set of discrete emotions $E = \{\text{anger, fear, joy, love, sadness, surprise}\}$.

This task presents several challenges typical of processing short, informal text. First, the limited length of the input results in sparse feature representations when using traditional bag-of-words methods. Second, the expression of emotion is often subtle and context-dependent, relying on sarcasm, negation, or idiomatic expressions that are difficult to capture without deep semantic understanding.

We approach this problem using the `dair-ai/emotion` dataset. The objective is to maximize the classification accuracy and the Macro-F1 score, the latter being particularly important to account for potential class imbalances among the six emotion categories. We contrast two distinct paradigms to solve this: a supervised approach relying on fixed embeddings and a zero-shot approach relying on natural language inference.

2 Methodology

To address the problem statement, we implement two parallel pipelines. Both leverage the Transformer architecture but utilize it in fundamentally different ways.

2.1 Supervised Learning: Feature Extraction

In the first approach, we treat a pretrained Transformer model as a feature extractor. The architecture consists of two components:

- **Backbone:** A pretrained model (e.g., RoBERTa or DistilBERT) from the `sentence-transformers` library. We process the input text x to obtain a dense vector representation $v \in \mathbb{R}^d$, where d is the embedding dimension. The weights of this backbone are frozen during training.

- **Classifier Head:** A lightweight supervised model (e.g., Logistic Regression or a Multi-Layer Perceptron) trained on the pair (v, y) . This head learns to map the semantic embedding to the specific emotion logits.

This method mirrors the "transfer learning" workflow common in computer vision, allowing us to leverage the massive pretraining of the Transformer while keeping the training cost low.

2.2 Zero-Shot Classification

The second approach utilizes a Zero-Shot classification paradigm based on Natural Language Inference (NLI). Instead of training a classifier on the specific emotion labels, we use a model pretrained on NLI tasks (determining if a hypothesis contradicts, entails, or is neutral to a premise).

For each input text x , we construct a set of hypotheses $H = \{h_1, \dots, h_6\}$ using a template such as "This text expresses $\langle emotion \rangle$ ". The model computes the entailment probability $P(h_i|x)$ for each emotion. The predicted label is the emotion corresponding to the hypothesis with the highest probability:

$$\hat{y} = \operatorname{argmax}_{e \in E} P(\text{"This text expresses } e \text{"} \mid x)$$

This method requires no training on the `dair-ai/emotion` dataset, testing the model's ability to generalize to unseen classes purely based on their semantic description.

Subsequent paragraphs, however, are indented.

Sample Heading (Third Level) Only two levels of headings should be numbered. Lower level headings remain unnumbered; they are formatted as run-in headings.

Sample Heading (Fourth Level) The contribution should contain no more than four levels of headings. Table 1 gives a summary of all heading levels.

Table 1. Table captions should be placed above the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Run-in Heading in Bold. Text follows	10 point, bold
4th-level heading	<i>Lowest Level Heading.</i> Text follows	10 point, italic

Displayed equations are centered and set on a separate line.

$$x + y = z \tag{1}$$

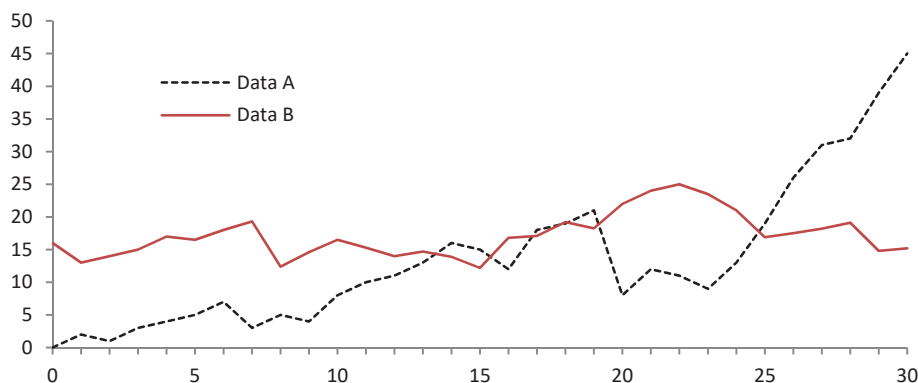


Fig. 1. A figure caption is always placed below the illustration. Please note that short captions are centered, while long ones are justified by the macro package automatically.

Please try to avoid rasterized images for line-art diagrams and schemas. Whenever possible, use vector graphics instead (see Fig. 1).

Theorem 1. *This is a sample theorem. The run-in heading is set in bold, while the following text appears in italics. Definitions, lemmas, propositions, and corollaries are styled the same way.*

Proof. Proofs, examples, and remarks have the initial word in italics, while the following text appears in normal font.

For citations of references, we prefer the use of square brackets and consecutive numbers. Citations using labels or the author/year convention are also acceptable. The following bibliography provides a sample reference list with entries for journal articles [1], an LNCS chapter [2], a book [3], proceedings without editors [4], and a homepage [5]. Multiple citations are grouped [1–3], [1, 3–5].

Acknowledgments. A bold run-in heading in small font size at the end of the paper is used for general acknowledgments, for example: This study was funded by X (grant number Y).

Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

¹ If EquinOCS, our proceedings submission system, is used, then the disclaimer can be provided directly in the system.

References

1. Author, F.: Article title. *Journal* **2**(5), 99–110 (2016)
2. Author, F., Author, S.: Title of a proceedings paper. In: Editor, F., Editor, S. (eds.) *CONFERENCE 2016, LNCS*, vol. 9999, pp. 1–13. Springer, Heidelberg (2016). <https://doi.org/10.10007/1234567890>
3. Author, F., Author, S., Author, T.: Book title. 2nd edn. Publisher, Location (1999)
4. Author, A.-B.: Contribution title. In: *9th International Proceedings on Proceedings*, pp. 1–2. Publisher, Location (2010)
5. LNCS Homepage, <http://www.springer.com/lncs>, last accessed 2023/10/25